Slide 7: the driver (THS4531)

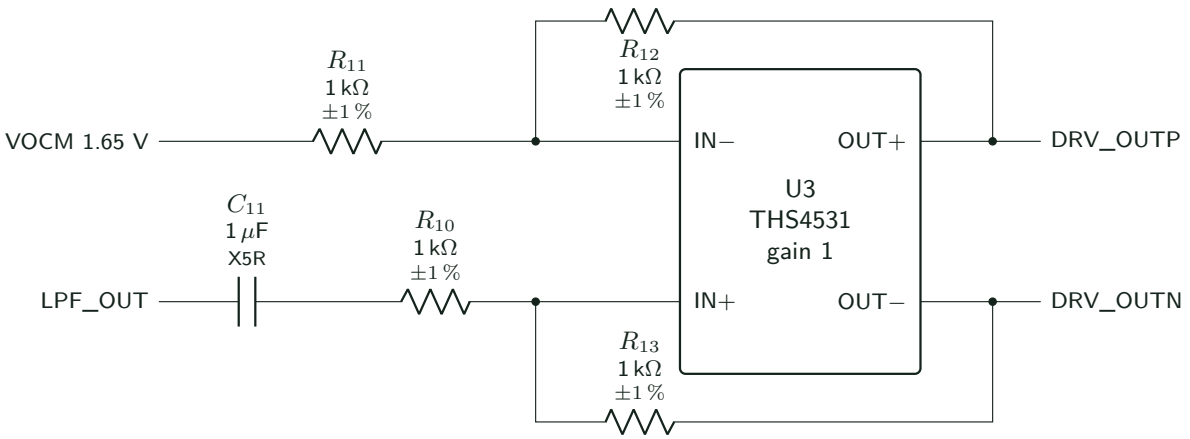
Turn one wire into the two the ADC needs

ADS7057 test board v3 — study notes

1 Why two wires

The ADS7057 measures the difference $V_{\mathrm{AINP}} - V_{\mathrm{AINM}}$, not a voltage to ground. Noise that reaches both inputs equally cancels. The low-pass output (LPF_OUT) is one wire. The driver U3 makes the second wire as a mirror image of the first, around the mid-rail VOCM = 1.65 V.

2 The circuit



All four resistors are 1 kΩ. C11 blocks DC. VOCM sets the centre of the two outputs (R14, R15 divide AVDD, C13 smooths it).

3 Derivation

An ideal fully differential amplifier keeps its two inputs at the same voltage, $v_P = v_N = v$, and forces the average of its outputs to VOCM:

$$\frac{v_{o+} + v_{o-}}{2} = V_{\mathrm{OCM}}.$$

Write the current balance at the two input nodes ($R = 1\,\mathrm{k\Omega}$ for all four resistors, no current enters the amplifier):

$$\text{node P:}\quad \frac{v_{\mathrm{in}} - v}{R} + \frac{v_{o-} - v}{R} = 0 \qquad \text{node N:}\quad \frac{V_{\mathrm{OCM}} - v}{R} + \frac{v_{o+} - v}{R} = 0.$$

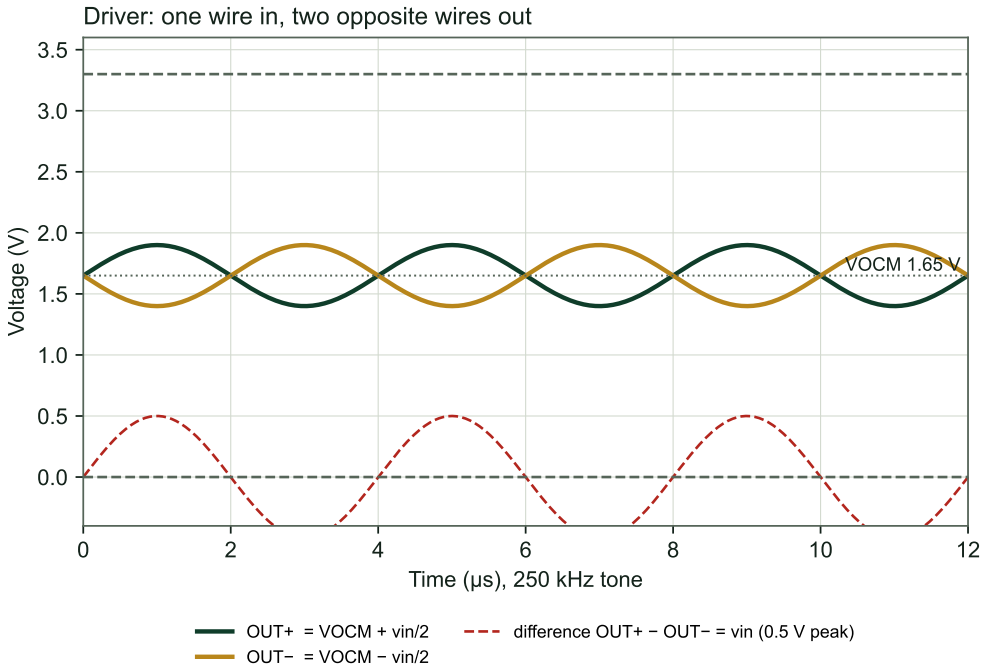
Subtract the second equation from the first. The v terms cancel:

$$v_{\mathrm{in}} - V_{\mathrm{OCM}} + v_{o-} - v_{o+} = 0 \;\Rightarrow\; \boxed{v_{o+} - v_{o-} = v_{\mathrm{in}} - V_{\mathrm{OCM}}}$$

For the signal alone (deviation from DC, so V_{OCM} does not move) the differential output equals the input: the gain is $R_{13}/R_{10} = 1$. With the common-mode condition each output carries half:

$$v_{o\pm} = V_{\mathrm{OCM}} \pm \frac{v_{\mathrm{in}}}{2}.$$

DC. C11 is open at DC, so only R13 and R11 act. Both outputs sit at V_{OCM} (ngspice: DRV_OUTP and DRV_OUTN at 1.64 V). Nothing is amplified at DC and nothing clips.



4 The input corner

The signal source sees C11 in series with the amplifier's input resistance. The two feedback paths act on both input nodes, so the input resistance is not R10 alone. Solve node P with $v_{o+} = 2v$, $v_{o-} = -2v$ (ac, outputs in opposition): $v = v_{\mathrm{in}}/4$, so

$$i_{\mathrm{in}} = \frac{v_{\mathrm{in}} - v}{R} = \frac{3}{4}\frac{v_{\mathrm{in}}}{R} \;\Rightarrow\; R_{\mathrm{in}} = \frac{4R}{3} = 1.33\,\mathrm{k\Omega}, \qquad f_c = \frac{1}{2\pi R_{\mathrm{in}} C_{11}} = 119\,\mathrm{Hz}.$$

Correction to the manual (step 17): it gives 159 Hz, which uses R10 alone. With the feedback included the corner is 119 Hz. Both are far below the 100 kHz band, so nothing else changes.

5 Numbers to remember

Item	Value
Supply and VOCM range (3.3 V)	2.5 to 5.5 V; VOCM 0.95 to 2.3 V
Gain, band	differential gain 1; bandwidth far above the band
Event of 120 mV at the head	about 0.5 V between AINP and AINM
Load between the outputs	400 pF behind 20 Ω, the datasheet suggests about 2 pF (open risk, slide 13)